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Neuroimaging abnormalities in functional/dissociative seizures: a meta-analysis

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ABSTRACT

The neural basis of functional/dissociative seizures represents a significant challenge for psychiatry and neurology. While abnormalities in neuroimaging have been observed, findings across studies remain inconsistent. This systematic review and meta-analyses examined neuroimaging abnormalities in functional/dissociative seizures, conducting separate meta-analyses of cortical thickness and whole-brain, seed-based, resting-state functional connectivity. A total of 22 studies were included in the review, of which seven and three were suitable for meta-analysis of structural and functional MRI data, respectively. Meta-analysis of cortical thickness from seven structural MRI studies (198 functional/dissociative seizures and 254 healthy controls) revealed a reduction in the right calcarine fissure/surrounding cortex ($z = -3.08$, $p < 0.01$) and right precentral gyrus ($z = -3.112$, $p < 0.01$). Additionally, the whole-brain seed-based resting-state functional connectivity analysis from three functional MRI studies (48 patients with functional/dissociative seizures and 64 healthy controls) demonstrated increased connectivity between seeds in the sensorimotor network (including posterior insula, paracentral lobule, postcentral gyrus, supplementary motor area and superior temporal gyrus) and areas of the ventral attention network, including left thalamus ($z = 3.73$, $p < 0.01$) and left putamen ($z = 3.62$, $p < 0.01$); and seeds in the default mode network (including ventral anterior insula, superior frontal gyrus and middle temporal gyrus) exhibited heightened connectivity with the left insula ($z = 4.01$, $p < 0.01$). These findings indicate structural and functional abnormalities in visual, sensorimotor, default mode, and attentional areas and networks associated with functional/dissociative seizures, highlighting potential neurobiological mechanisms. Further research is necessary to elucidate the complex interplay of clinical features and comorbidities, which will enhance our understanding of the pathophysiology of functional/dissociative seizures.

1. Introduction

Functional Neurological Disorder (FND) is a heterogeneous condition characterized by symptoms not explained by structural neurological disease, including motor disturbances, sensory changes, and cognitive or speech impairments (Hallett et al., 2022a). Among its subtypes, functional/dissociative seizures (FDS), also known as psychogenic non-epileptic seizures, are among the most prevalent and clinically significant presentations. Historically, FDS has been understood to arise in grossly normal brains, often conceptualized within psychodynamic or behavioral frameworks. With advances in MRI, however, studies began to demonstrate subtle abnormalities in both brain structure and connectivity, leading to a paradigm shift from viewing FDS as a disorder without biological underpinnings to one increasingly explained by disruptions in brain network function. Findings across studies remain inconsistent, however, and reliable neurobiological markers remain elusive (Mcsweeney et al., 2017).

Since Reuber *et al.* suggested a potential association between structural brain abnormalities and the occurrence of FDS events (Reuber et al., 2002), structural neuroimaging studies of FDS have yielded surprising and diverse results. On the other hand, the term “functional” implies a condition wherein the fundamental pathophysiological

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mechanisms are primarily in the functioning of brain networks rather than alteration in brain structures (Hallett et al., 2022b). This has led to an increased interest in brain network function and in communication between large-scale brain networks in the pathophysiology of FS, particularly using resting-state functional connectivity (rsFC). Although initial findings support the notion that FDS is associated with abnormal rsFC, inconsistencies in the location and direction of effects pose challenges to unifying this research (Mcsweeney et al., 2017; Bègue et al., 2019; Foroughi et al., 2020).

A variety of factors may contribute to the challenges in identifying neuroimaging marker(s) in FS. Firstly, FDS appears not to be a unitary disease; it is a heterogeneous syndrome, encompassing diverse, co-occurring symptoms with potentially distinct neurobiological underpinnings (Magaudda et al., 2016). Secondly, research studies in the field have been limited by small sample sizes, reducing the likelihood of uncovering true effects and inflating effect size estimates when a true effect is identified (Button et al., 2013). Thirdly, the use of varied analysis techniques, including differences in pre-processing protocols and region-of-interest (ROI) analyses, adds another layer of complexity. For instance, prior research using seed-based rsFC show considerable variability in the selection of seed ROIs. All these factors complicate replication efforts, contributing to the inconsistency and poor reproducibility of neuroimaging results in FS.

To address the heterogeneity and enhance reproducibility in FDS neuroimaging findings, one approach is the meta-analysis of smaller studies. While several reviews have examined structural and functional abnormalities associated with FDS (Mcsweeney et al., 2017; Bègue et al., 2019; Foroughi et al., 2020), only one meta-analysis has been published (Mcsweeney et al., 2017). McSweeney et al. conducted three separate coordinate-based Activation Likelihood Estimation (ALE) analyses across nine eligible studies (307 participants). The first analysis, which pooled both structural and functional imaging studies, and the second, which focused exclusively on functional connectivity studies (six studies, 141 participants), both yielded no significant clusters. Only the third analysis, restricted to three structural MRI studies (166 participants), identified significant effects, revealing a single cluster in the left temporal lobe (Brodmann area 21). However, in that study, data from diverse imaging modalities and analysis techniques were pooled together, hindering the ability to extract reliable results across studies. Furthermore, no meta-analysis has been reported on the growing body of whole-brain seed-based rsFC studies.

This review aims to fill the current knowledge gap by systematically analysing quantitative structural and functional MRI studies that compared patients with FDS to healthy individuals, and providing separate meta-analyses for structural and functional MRI studies where data permit. The aim was to identify spatially consistent brain regions across studies, shedding light on potential neural mechanisms of FS.

2. Method

2.1. Search strategy and selection criteria

In accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Moher et al., 2009; Shamseer et al., 2015), we performed a systematic review and meta-analysis, which was pre-registered with the International Prospective Register of Systematic Reviews (www.crd.york.ac.uk/prospero) (ID=CRD42024526975). A comprehensive and systematic literature search was conducted in MEDLINE, Embase, PsycINFO and Web of Science electronic databases on September 28, 2023, and updated on January 1, 2025. The search terms used are detailed in the Supplementary Material. The reference lists of reviewed studies and relevant literature reviews were examined to identify any additional publications. We included studies that: 1) compared adult patients with a diagnosis of pure FDS to healthy controls (HC); 2) investigated quantitative structural or functional MRI in participants; and 3) were published in English. Conference abstracts, case reports, reviews, case series and editorials were excluded; no limitations were set on the publication date. It should be noted that studies involving mixed FND cohorts or patients with FDS and comorbid epilepsy were excluded, unless results for the pure FDS subgroup were reported separately; in such cases, only the FDS-specific data were extracted for analysis.

The first two authors independently screened records for eligibility based on the established criteria. The screening process began with an evaluation of titles and abstracts, followed by a full-text review of potentially eligible studies. Any disagreements were addressed through discussion; if consensus could not be reached, a third reviewer (RK) was involved to make a final decision.

2.2. Quality assessment

The risk of methodological bias in individual studies was assessed using a customized version of the National Heart, Lung, and Blood Institute's quality assessment tool for case-control studies by MH and AL to align with our target studies (Study Quality Assessment Tools, 2024). Key domains such as clarity of research question/objective, study population, case/control selection, comparability of cases and controls, the assessment of neuroimaging techniques, and confounding variables such as age, sex, illness duration and seizure/episode frequency, psychiatric/neurological comorbidities, and medication use were considered to pose a risk of bias. Based on these domains, each study's risk of bias was classified as "good," "fair," or "poor" (see Table 1).

2.3. Data extraction and preparation for meta-analyses

Data synthesis was performed as a narrative report in the form of tables, providing an overview of demographic and clinical characteristics, basic methodological details and scanning parameters. The differences between FDS and HC for each modality were presented in colour-coded tables. We conducted separate meta-analyses for the two most prevalent measures, cortical thickness and whole-brain seed-based rsFC.

For the meta-analysis of cortical thickness, we utilized seed-based d mapping (SDM) software (version 6.23) (www.sdmproject.com). Based on cluster peak coordinates and effect sizes of significant group differences, this software creates an effect-size signed map for each study, employing an anisotropic Gaussian kernel (Radua and Mataix-Cols, 2009; Radua et al., 2014). To ensure a focused analysis of cortical regions and enable surface-based morphometric analysis, we employed the FreeSurfer mask (http://surfer.nmr.mgh.harvard.edu). Positive and negative effect coordinates were reconstructed within the same map to ensure that no voxel represented opposing directions simultaneously (Radua and Mataix-Cols, 2009). The individual maps were then combined through meta-analytic calculations, with weights assigned based on intra-study variance and inter-study heterogeneity. Significant clusters were defined using standard permutation tests, with a primary threshold of uncorrected p-value < 0.005 and $|z| > 1$. This threshold approximates a corrected p-value of < 0.05 in SDM based on empirical comparisons (Radua and Mataix-Cols, 2012).

To address the variability in locations and sizes of seed ROIs employed (Fox, 2010; Power et al., 2012), we adopted an innovative meta-analytic strategy (Kaiser et al., 2015). This approach categorizes seeds into a predefined functional network parcellations according to their locations, allowing for the combination of studies with seeds situated within the same network. Coordinates for each seed ROI were collected, and peak coordinates for each between-group effect demonstrating significance at the whole-brain level were identified. Each seed ROI was categorized into a "seed/ROI-network" according to its location within rsFC networks, defined by a comprehensive whole-brain network parcellation (Choi et al., 2012a; Schaefer et al., 2018), which includes the visual (VN), sensorimotor (SMN), dorsal attention (DAN), ventral attention (VAN), limbic (LN), frontoparietal (FPN), and default mode network (DMN). Peak effects were categorized into hyper-connectivity

Table 1
Demographic and clinical characteristics of FDS and HC participants across included studies.

Study	Number (female)		Age (years ± SD)		Average age at FDS onset (years ± SD)	Average FDS duration (years ± SD)	Average FDS frequency (per month)	VEEG-confirmed FDS diagnosis	Comorbid epilepsy	Handedness	Quality check
	FDS	HC	FDS	HC							
Labate *et al.* 2012	20 (11)	40 (21)	37±14	36±10	23±8	N/A	N/A	yes	none	N/A	fair
van der Kruijs *et al.* 2012	11 (6)	12 (8)	34±11	34±11	N/A	8±8	2±3	N/A	none	N/A	fair
Ding *et al.* 2013	17(10)	20 (12)	20±8	22±2	N/A	3±5	N/A	yes	none	N/A	fair
Ding *et al.* 2014	18 (11)	20 (12)	18±5	22±2	N/A	1±2	N/A	yes	none	N/A	fair
van der Kruijs *et al.* 2014	21 (13)	27 (21)	34±12	36±12	N/A	6±7	12±34	N/A	none	N/A	fair
Li, *et al.* 2015a	18 (11)	20 (12)	18±5	22±2	N/A	1±2	3±2	yes	none	N/A	fair
Li, *et al.* 2015b	18 (11)	20 (12)	18±5	22±2	N/A	1±2	3±2	yes	none	N/A	fair
Lee *et al.* 2015	16 (15)	16 (15)	40±14	40±14	N/A	N/A	N/A	yes	none	N/A	fair
Ristić *et al.* 2015	37 (31)	37 (26)	37±14	38±13	26±11	11±11	#	yes	none	94.5% RH	fair
Mcsweeney *et al.* 2018	20 (14)	20 (14)	41±13	41±12	28±12	10±14	N/A	yes	none	100% RH	fair
Vasta *et al.* 2018	23 (20)	21 (15)	26±12	29±8	N/A	16±7	N/A	yes	none	N/A	fair
Szaflarski *et al.* 2018	12 (10)	24 (20)	36±12	36±11	32±12	4±4	8 (1–60)**	yes	none	N/A	fair
Sone *et al.* 2019	17 (13)	26 (20)	29±9	30±7	22±10	6±6	N/A	yes	none	N/A	fair
Allendorfer *et al.* 2019	12 (11)	12 (11)	45 (21)*	36 (15)*	37 (18)*	2 (3)*	24 (28)*+	yes	none	N/A	fair
Tatekawa *et al.* 2020	29 (23)	29 (23)	38 (34–43)**	46 (43-50)**	N/A	11 (7–16)**	N/A	yes	none	N/A	fair
Amiri *et al.* 2021a	23 (14)	25 (15)	30±11	32±6	N/A	1±1	5±6	yes	none	100% RH	fair
Amiri, *et al.* 2021b	23 (14)	25 (15)	30±11	32±6	N/A	1±1	5±6	yes	none	100% RH	fair
Labate *et al.* 2021	42 (33)	78 (41)	42±14	40±11	31±11	N/A	N/A	yes	none	N/A	fair
Baykara *et al.* 2022	20 (20)	20 (20)	33±10	38±11	N/A	N/A	N/A	N/A	none	N/A	fair
Baykara *et al.* 2022	20 (20)	20 (20)	34±10	33±9	N/A	N/A	N/A	N/A	none	N/A	fair
Nasrullah *et al.* 2023	37 (37)	37 (37)	34±10	34±10	N/A	N/A	N/A	yes	none	100% RH	fair
Tsalouchidou *et al.* 2023	21 (14)	21 (14)	24 (17–44)*	24 (19–38)*	23 (13–43)*	2 (1–18)*	N/A	yes	none	N/A	fair

[*] data reported as median (inter-quartile range); [+] number of seizures is reported in the past 3 months; ** data reported in the parenthesis is the range of the parameter; # Seizure frequency distribution: 5.4 % report less than 1/year, 8.1 % report 1/year to 1/month, 37.8 % report 1/month to 1/week, and 48.6 % report more than 1/week. Abbreviations: FDS, Functional/Dissociative Seizures; N/A, Not Available; RH, Right-handed; SD, Standard Deviation; VEEG, Video Electroencephalography.

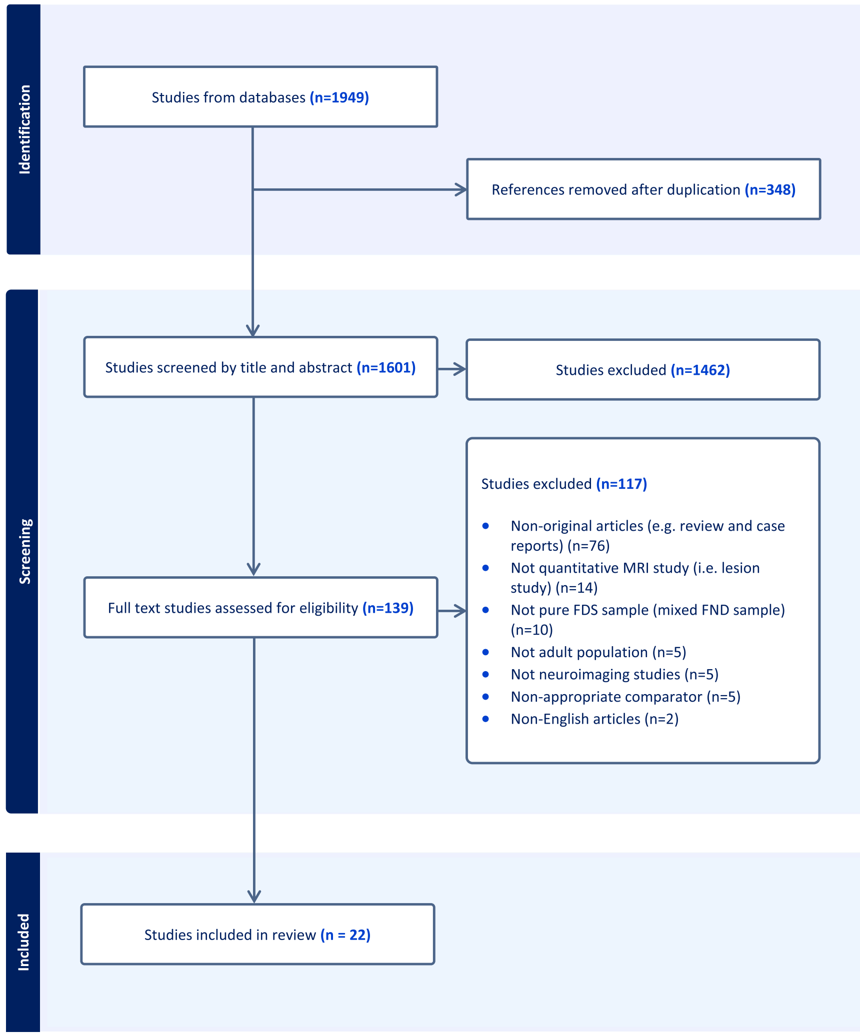
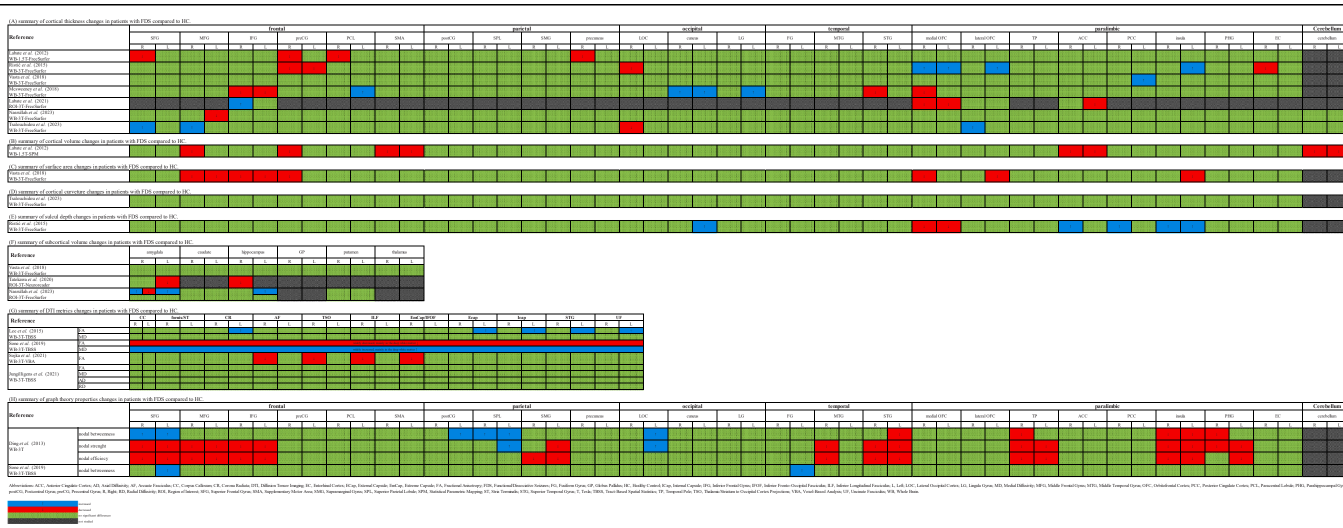


Fig. 1. PRISMA flowchart illustrating the systematic review inclusion process, along with an explanation of the criteria for excluded studies.

Table 2
Summary of structural MRI changes in patients with FDS compared to HC.



Abbreviations: ACC, Anterior Cingulate Cortex; AD, Axial Diffusivity; AF, Arcuate Fasciculus; CC, Corpus Callosum; CR, Corona Radiata; DTI, Diffusion Tensor Imaging; EC, Entorhinal Cortex; ECap, External Capsule; EmCap, Extreme Capsule; FA, Fractional Anisotropy; FDS, Functional/Dissociative Seizures; FG, Fusiform Gyrus; GP, Globus Pallidus; HC, Healthy Control; ICap, Internal Capsule; IFG, Inferior Frontal Gyrus; IFOF, Inferior Fronto-Occipital Fasciculus; ILF, Inferior Longitudinal Fasciculus; L, Left; LOC, Lateral Occipital Cortex; LG, Lingula Gyrus; MD, Medial Diffusivity; MFG, Middle Frontal Gyrus; MTG, Middle Temporal Gyrus; OFC, Orbitofrontal Cortex; PCC, Posterior Cingulate Cortex; PCL, Paracentral Lobule; PHG, Parahippocampal Gyrus; postCG, Postcentral Gyrus; preCG, Precentral Gyrus; R, Right; RD, Radial Diffusivity; ROI, Region of Interest; SFG, Superior Frontal Gyrus; SMA, Supplementary Motor Area; SMG, Supramarginal Gyrus; SPL, Superior Parietal Lobule; SPM, Statistical Parametric Mapping; ST, Stria Terminalis; STG, Superior Temporal Gyrus; T, Tesla; TBSS, Tract-Based Spatial Statistics; TP, Temporal Pole; TSO, Thalamic/Striatum to Occipital Cortex Projections; VBA, Voxel-Based Analysis; UF, Uncinate Fasciculus; WB, Whole Brain.

Table 3
Summary of functional MRI changes in patients with FDS compared to HC.

(A) Summary of whole-brain seed-based rsFC changes in patients with FDS compared to HC.

Reference	Clusters	Seed ROI-Network: VN	SMN					DAN		VAN	LN		FPN		DMN		
		precuneus	insula-P	PCL	postCG	SMA	STG	IPS	preCS	insula-DA	amygdala	PHG	IFG	SMG	insula-VA	SFG	MTG
van der Kruijs *et al.* (2012) WB-3T								R PCC ↑	R ACC ↑				R PCC ↑	R PCC ↑			
								R PCC ↑	R insula-A ↑				R ACC ↑	R insula-P ↑			
								R ACC ↑	R insula-P ↑				R CS ↑	R POF ↑			
								R ACC ↑	R CS ↑				R POF ↑				
								R insula-P ↑	R POF ↑								
								R insula-P ↑									
								R CS ↑									
								R CS ↑									
								R POF ↑									
								R POF ↑									
Li *et al.* (2015a) WB-3T		L SFG ↑		L MFG ↑	L SPL ↑	R ACC ↑							R MFG ↓			L IFG ↑	
				R SMA ↑	L MFG ↑	L MFG ↑										R PCL ↑	
Li *et al.* (2015b) WB-3T			R putamen ↑							L SPL ↑					R SMA ↑		
															L SMA ↑		
															L postCG ↑		
															R LG ↑		
Szaflarski *et al.* (2018) WB-3T							R precuneus ↑					L precuneus ↑					L MTG ↑
							L precuneus ↑					L insula ↑					R EC ↑
							R precuneus ↑					L TP ↑					L FO ↑
							L thalamus ↑					R ITG ↑					L MFG ↑
							R SMG ↑					R insula-A ↑					R caudate ↑
							R FO ↑					R insula ↑					L LG ↑
							R ITG ↑					R putamen ↑					
							L FG ↑					L LG ↑					
							R FG ↑					L FG ↑					
							R SMG ↑										
Allendorfer *et al.* (2019) WB-3T											L IFG/MFG ↑						
											L preCG ↑						

(B) summary of fALFF changes in patients with FDS compared to HC.

Reference	Functional Network: VN				SMN						DAN	VAN		LN	FPN								DMN	
Li *et al.* (2015) WB-3T	L precuneus ↑					R postCG ↑	L PCL ↑	L SMA ↑										R IFG ↓					L SFG ↑	

(C) summary of graph theory properties changes in patients with FDS compared to HC.

Amiri *et al.* (2021) WB-3T		R MOG ↓					R PCL ↑					B insula ↓	R putamen ↓						R caudate ↑					L IFG ↑

(D) summary of FCD changes in patients with FDS compared to HC.

Ding *et al.* (2014) WB-3T	long-range FCD			B CF ↑	B LG ↑	R preCG ↑	R postCG ↑	L PCl ↑	B SMA ↑	R STG ↑						R SMG ↓	R medial PFC ↓	R IPL ↓	R IFG ↓			R SFG ↓	R MFG ↓		
	short-range FCD										B MCC ↑										L SMA ↑	L SFG ↑	L MFG ↑		

ACC, Anterior Cingulate Cortex; B, Bilateral; CF, Calcarine Fissure; CS, Central Sulcus; DAN, Dorsal Attention Network; DMN, Default Mode Network; EC, Entorhinal Cortex; fALFF, fractional amplitude of low-frequency fluctuations; FCD, Functional Connectivity Density; FDS, Functional/Dissociative Seizures; FG, Fusiform Gyrus; FO, Frontal Operculum; FPN, Frontoparietal Network; HC, Healthy Control; IFG, Inferior Frontal Gyrus; insula-A, Anterior Part of Insula; insula-DA, Dorsal Anterior Part of Insula; insula-P, Posterior Part of Insula; insula-VA, Ventral Anterior Part of Insula; IPL, Inferior Parietal Lobule; IPS, Intraparietal Sulcus; ITG, Inferior Temporal Gyrus; L, Left; LG, Lingula Gyrus; LN, Limbic Network; MCC, Middle Cingulate Cortex; MFG, Middle Frontal Gyrus; MOG, Middle Occipital Gyrus; MTG, Middle Temporal Gyrus; PCC, Posterior Cingulate Cortex; PCL, Paracentral Lobule; PFC, Prefrontal Cortex; PHG, Parahippocampal Gyrus; POF, Parietal Occipital Fissure; postCG, Postcentral Gyrus; preCG, Precentral Gyrus; preCS, Precentral Sulcus; R, Right; rsFC, resting-state functional connectivity; SFG, Superior Frontal Gyrus; SMA, Supplementary Motor Area; SMG, Supramarginal Gyrus; SMN, Somatosensory Motor Network; STG, Superior Temporal Gyrus; T, Tesla; TP, Temporal Pole; VAN, Ventral Attention Network; VN, Visual Network; WB, Whole Brain.

↑ increased
↓ decreased
no significant differences
not studied

Table 4
Meta-analysis of abnormal cortical thickness in patients with FDS compared to HC (voxel threshold $p < 0.005$ and $|z| > 1$).

MNI coordinate	SDM-*z*	*p*-value	Voxels	Description
14,−98,−2	−3.08	< 0.01	14	right calcarine fissure/surrounding cortex
34,−18,62	−3.11	< 0.01	8	right precentral gyrus

Abbreviations: FDS, Functional/Dissociative Seizures; HC, Healthy Control; MNI, Montreal Neurological Institute; SDM, Seed-based d Mapping.

or hypo-connectivity based on the direction of the effect. Hyper-connectivity was defined as increased rsFC (greater positivity or reduced negativity of connectivity) in FDS patients compared to HC, while hypo-connectivity was defined as decreased rsFC (lower positivity or higher negativity of connectivity) in FDS patients relative to HC. Finally, we employed Ginger ALE/2.3.6 (https://www.brainmap.org/ale) to assess the functional convergence of differences among whole-brain seed/ROI-network-based rsFC neuroimaging research in FS. The ALE method extracted coordinates from the included studies to test for anatomical consistency and concordance. Coordinates were weighted based on the sample size, contributing to estimates of anatomical likelihood estimation for each intracerebral voxel on a standardized map (Eickhoff et al., 2017). The statistical significance of the ALE scores was determined by a permutation test (1000 iterations), applying cluster-level inference at p-value < 0.05 (FWE), with a cluster-forming threshold set at p-value < 0.05. Moreover, we assessed sensitivity by quantifying the number of studies contributing to each significant cluster.

3. Results

A total of 1949 publications were retrieved. After removal of duplicates, 1601 studies were screened for eligibility based on their titles and abstracts. This resulted in 139 studies for full-text review. Ultimately, we identified 22 original articles meeting our inclusion criteria, comprising 15 structural MRI and seven functional MRI studies (see Fig. 1).

Comprehensive details of the included studies and their quality assessments are summarized in Table 1. Outcomes from studies using different analytical methods are reported separately and summarised in color-coded Table 2A-H and Table 3A-D. The meta-analytic results for cortical thickness studies are summarized in Table 4, and those for whole-brain seed-based rsFC in Table 5.

3.1. Structural MRI

Fifteen studies investigated structural changes in FS. Among them, eight studies concentrated on morphometric analysis, encompassing measurements such as cortical thickness (Labate et al., 2012, 2021a; Ristić et al., 2015; Mcsweeney et al., 2018; Vasta et al., 2018a; Nasrullah

Table 5
Meta-analysis of abnormal rsFC of SMN and DMN seed-network in patients with FDS compared to HC (a permutation test with 1000 iterations, utilizing cluster-level inference at $p < 0.05$, and a cluster-forming threshold set at $p < 0.05$).

(A) SMN seed-network

Cluster number	MNI coordinate	ALE	*p*-value	ALE-*z*	Description
1	−11,−18,8	0.01	< 0.01	3.73	left thalamus (VAN)
1	−27,0,9	0.01	< 0.01	3.62	left putamen (VAN)

(B) DMN seed-network

Cluster number	MNI coordinate	ALE	*p*-value	ALE-*z*	Description
1	−48,−24,26	0.01	< 0.01	4.01	left insula (SMN)
1	−47,−8,14	0.01	< 0.01	3.95	left insula (SMN)

Abbreviations: ALE, Activation Likelihood Estimation; DMN, Default Mode Network; FDS, Functional/Dissociative Seizures; HC, Healthy Control; MNI, Montreal Neurological Institute; rsFC, resting-state functional connectivity; SMN, Sensorimotor Network; VAN, Ventral Attention Network.

et al., 2023; Tsalouchidou et al., 2023), cortical volume (Labate et al., 2012), surface area (Vasta et al., 2018a), cortical curvature (Tsalouchidou et al., 2023), sulcal depth (Ristić et al., 2015), and subcortical volume (Vasta et al., 2018a; Nasrullah et al., 2023; Tatekawa et al., 2020); one study evaluated textural parameters (Baykara and Baykara, 2022); one study assessed shape deformations (Baykara et al., 2022) and five studies investigated structural connectivity utilising diffusion tensor imaging (DTI) (Sone et al., 2019a; Ding et al., 2013) and graph theory analysis (Lee et al., 2015a; Sone et al., 2019b; Sojka et al., 2021a; Jungilligens et al., 2021).

3.2. Morphometric analysis

Eight studies employed brain morphometry modalities using two primary techniques: voxel-based morphometry (VBM) and surface-based analysis (SBA). The most frequently utilized method was SBA via FreeSurfer, employed in measuring cortical thickness ($n = 7$). Six of these utilized whole-brain approaches (Labate et al., 2012; Ristić et al., 2015; Mcsweeney et al., 2018; Vasta et al., 2018a), while one study opted for an ROI approach (Labate et al., 2021b). FreeSurfer was also the chosen method for analysing surface area ($n = 1$) (Vasta et al., 2018b), cortical curvature ($n = 1$) (Tsalouchidou et al., 2023), and sulcal depth ($n = 1$) (Ristić et al., 2015). Additionally, one study employed VBM to assess cortical volume (Labate et al., 2012). For subcortical volumes, one study measured subcortical volume using SBA and FreeSurfer in a whole-brain approach (Vasta et al., 2018b), while two studies employed an ROI approach, using SBA via FreeSurfer (Nasrullah et al., 2023) or VBM via Neuroreader (Tatekawa et al., 2020).

3.2.1. Cortical thickness

Pooling the seven cortical thickness studies involved a total of 455 participants (201 FDS and 258 HC), revealing differences in cortical thickness across spatially heterogeneous regions. These differences included both increases and decreases in cortical thickness, predominantly affecting frontal (bilateral precentral gyrus, bilateral inferior frontal gyrus, bilateral paracentral lobule, bilateral middle frontal gyrus and right superior frontal gyrus) and paralimbic regions (including bilateral medial orbitofrontal cortex, bilateral lateral orbitofrontal cortex, left anterior cingulate cortex, left posterior cingulate cortex, left insula and right entorhinal cortex). Other affected areas included the occipital region (bilateral cuneus, right lateral occipital gyrus, and left lingual gyrus), temporal lobe (right superior temporal gyrus), and parietal area (right precuneus) (Labate et al., 2012, 2021a; Ristić et al., 2015; Mcsweeney et al., 2018; Vasta et al., 2018a; Nasrullah et al., 2023; Tsalouchidou et al., 2023). Detailed results are summarized in Table 2A. In summary, cortical thickness abnormalities in FDS were most prominent in frontal, paralimbic, and occipital areas, with less involvement of temporal and parietal regions.

3.2.2. Cortical volume

Cortical volume in FDS patients was investigated in one study (20 FDS and 40 HC), revealing a reduction in volume within frontal regions (right middle frontal gyrus and right precentral gyri and bilateral supplementary motor area), paralimbic areas (bilateral anterior cingulate cortex) and bilateral cerebellum compared to HC (Labate et al., 2012) (Table 2B).

3.2.3. Surface area

A single study investigated surface area abnormalities in FDS individuals (23 FDS and 21 HC), revealing reduced surface area in frontal (bilateral and inferior frontal gyri, and right precentral gyrus) and paralimbic (right medial orbitofrontal and left lateral orbitofrontal cortices and left insula) regions compared to HC (Vasta et al., 2018b) (Table 2C).

3.2.4. Cortical curvature

One study examined cortical curvature abnormalities in FDS individuals (21 FDS and 21 HC), finding no significant differences from HC (Tsalouchidou et al., 2023) (Table 2D).

3.2.5. Sulcal depth

Sulcal depth was assessed in a single study (37 FDS and 37 HC), reporting both decreases and increases within paralimbic areas (decreases in bilateral medial orbitofrontal cortex and increases in right anterior and right posterior cingulate cortex and bilateral insula) and increased sulcal depth in occipital region (cuneus) compared to HC (Ristić et al., 2015) (Table 2E).

3.2.6. Subcortical volume

Subcortical volume was assessed in three studies (89 FDS and 87 HC). One study utilized a whole-brain approach and found no significant differences between FDS and HC (Vasta et al., 2018a). The second study employed an ROI approach, focusing on the amygdala and hippocampus, reporting decreased volume in limbic areas (left amygdala and right hippocampus) in FDS compared to HC (Tatekawa et al., 2020). The third study also used an ROI approach, targeting the amygdala, hippocampus, caudate, and putamen, but found no significant changes between FDS and HC. However, within the subfields of the amygdala and hippocampus, they observed increases in the left dentate gyrus of the hippocampus, the left anterior amygdala, the right central and medial amygdala, and a decrease in the right lateral amygdala (Nasrullah et al., 2023), see Table 2F. In summary, findings on subcortical volume in FDS were inconsistent and largely based on ROI analyses, with only limited and variable evidence for abnormalities in limbic structures such as the amygdala and hippocampus.

3.3. Texture analysis

One study used texture analysis to evaluate the dorsal striatum nuclei, including caudate and putamen, in 20 individuals with FDS compared to 20 HC (Baykara and Baykara, 2022). Texture analysis is a post-processing technique that quantifies the variation in pixel intensity patterns using descriptive statistics to assess tissue characteristics (Somwanshi et al., 2017). The results showed increased textural parameters for the putamen and caudate nucleus, in FDS compared to HC.

3.4. Statistical shape analysis

One study utilized statistical shape analysis to identify deformations in the corpus callosum between 20 FDS subjects and 20 HC (Baykara et al., 2022). The analysis revealed significant shape differences in the corpus callosum, with the maximum deformations observed in the posterior region. However, there was no significant difference in the corpus callosum area between the groups.

3.5. Structural connectivity

3.5.1. DTI

Four studies assessed diffusion parameters in FS. Among these, four studies focused on fractional anisotropy (FA) (Lee et al., 2015a; Sone et al., 2019b; Sojka et al., 2021a; Jungilligens et al., 2021), three studies on mean diffusivity (MD) (Lee et al., 2015a; Sone et al., 2019b; Jungilligens et al., 2021), and one on axial (AD) and radial diffusivity (RD) (Jungilligens et al., 2021). Notably, a tract-based spatial statistics (TBSS) study identified increased FA in the left corona radiata, internal capsule, external capsule, uncinate fasciculus, and superior temporal gyrus (Lee et al., 2015b), while another TBSS study observed widespread FA increases, primarily in deep white matter regions (Sone et al., 2019b). However, the third TBSS analysis found no significant differences in FA (Jungilligens et al., 2021). In contrast, the only voxel-based analysis reported reduced FA in the right hemisphere, affecting the arcuate fasciculus, thalamic/striatum to occipital cortex projections, inferior longitudinal fasciculus, and extreme capsule/ inferior fronto-occipital fasciculus (Sojka et al., 2021b). In terms of MD, one TBSS study noted widespread increases, mainly in deep white matter (Sone et al., 2019b), while the other two reported no differences between FDS and HC (Lee et al., 2015a; Jungilligens et al., 2021). Additionally, the sole study investigating AD and RD found no significant differences in FDS compared to HC (Jungilligens et al., 2021) using TBSS. Detailed results are summarized in Table 2G. In summary, DTI studies in FDS reported heterogeneous and partly conflicting findings, with some showing widespread white matter differences and others finding no significant abnormalities.

3.5.2. Graph theory properties

Two studies investigated abnormalities in nodal characteristics within the whole-brain network of individuals with FDS using DTI. One study exclusively measured nodal betweenness, revealing increased values in the frontal lobe (left superior frontal gyrus) and the temporal lobe (left fusiform gyrus) (Sone et al., 2019b). The other study reported more widespread changes in nodal betweenness in FS, with increases observed in frontal areas (bilateral superior frontal gyrus), parietal regions (left postcentral gyrus and bilateral superior parietal gyrus), and the occipital lobe (left lateral occipital cortex). Additionally, decreases in nodal betweenness were noted in paralimbic areas (right temporal pole, bilateral insula, and right parahippocampal gyrus) and the temporal region (left superior temporal gyrus) (Ding et al., 2013). Furthermore, this study explored nodal efficiency and strength as additional measures of network properties, revealing a similar pattern of reductions in the frontal lobe (including bilateral superior, middle and inferior frontal gyri), paralimbic regions (bilateral temporal pole, bilateral insula, and parahippocampal gyrus), temporal lobe (right middle temporal gyrus and bilateral superior temporal gyrus), and parietal lobe (supramarginal gyrus); see Table 2H.

3.6. Functional MRI

3.6.1. Resting-state functional MRI

In the eight resting-state fMRI studies, the most frequently used method of analysis was whole-brain seed-based rsFC analysis ($n = 4$) (Van Der Kruijs et al., 2012; Li et al., 2015a, 2015b; Allendorfer et al., 2019). Other analysis methods included fractional amplitude of low-frequency fluctuations (fALFF; $n = 1$) (Li et al., 2015a), graph theory properties ($n = 1$) (Amiri et al., 2021a), functional connectivity density (FCD) analysis ($n = 1$) (Ding et al., 2014), seed-to-seed rsFC ($n = 1$) (Amiri et al., 2021b).

3.6.1.1. Whole-brain seed-based rsFC. The four studies employing whole-brain seed-based rsFC analysis involved a total of 123 participants (59 individuals with FDS and 64 HC). In total, there were 45 connectivity pairs of significantly altered functional connectivity (FC) between seed ROIs and clusters. Interestingly, 44 were identified as higher in FDS compared to healthy individuals (indicating either increased positive or reduced negative rsFC in FDS patients compared to HC; i.e., FDS > HC), while one exhibited the opposite pattern (either increased negative or reduced positive rsFC in FDS patients compared with HC; i.e., FDS < HC).

Among the 44 connectivity pairs demonstrating higher FC in FS, the most frequently reported connectivity patterns involved interactions between SMN and FPN regions (10 increased connectivity across 3 studies). This was followed by SMN-DMN regions (5 increased connectivity across 2 studies), FPN-VN regions (3 increased connectivity across 2 studies), FPN-VA regions (2 increased connectivity across 2 studies) and DMN-DMN regions (2 increased connectivity across 2 studies); see Table 3A.

As a resting-state functional MRI signal, fractional amplitude of low-frequency fluctuation (fALFF), which encompasses the low-frequency power spectrum (0.01–0.08 Hz), reflects the intensity of local spontaneous brain activity by quantifying low-frequency oscillations. A voxel-wise fALFF analysis was conducted to assess regional neuronal activity at rest in FDS patients. Results indicated significant increases in fALFF within the sensorimotor network (SMN), specifically in the right post-central gyrus, left paracentral lobule, and left supplementary motor area. Additionally, elevated fALFF was observed in the default mode network (DMN) in the left superior frontal gyrus and in the visual network (VN) in the left precuneus. Conversely, decreased fALFF was identified in the frontoparietal network (FPN), specifically in the right inferior frontal gyrus pars triangularis. In summary, whole-brain seed-based rsFC studies in FDS consistently demonstrated increased connectivity, most prominently involving sensorimotor, default mode, dorsal attention, and limbic networks.

3.6.1.2. Fractional amplitude of low-frequency fluctuations. A voxel-wise fALFF analysis was conducted to investigate regional neuronal activity at rest in 18 FDS patients compared to 20 HC. The FDS patients exhibited notable increases in fALFF within the SMN (right postcentral gyrus, left paracentral lobule and left supplementary motor area). Moreover, elevated fALFF was observed in the DMN (left superior frontal gyrus) and VN (left precuneus). Conversely, decreased fALFF was noted in the FPN (right inferior frontal gyrus pars triangularis) (Li et al., 2015a); see Table 3B.

3.6.1.3. Graph theory properties. One study conducted graph theory analysis of functional MRI data on 23 FDS and 25 HC (Amiri et al., 2021a). This highlighted areas exhibiting hypo-connectivity (i.e., decreased nodal degree) primarily implicated in VAN (bilateral insula and right putamen), along with another region in the VN (right middle occipital gyrus). Moreover, it identified several areas of hyper-connectivity (i.e., increased nodal degree) in SMN (right paracentral lobule), DMN (left inferior frontal gyrus pars orbitalis) and FPN (right caudate); see Table 3C.

3.6.1.4. Functional connectivity density. Findings from the only FCD study on 18 FDS and 20 HC primarily revealed increased long-range FCD in SMN (left paracentral lobule, right postcentral gyrus, right precentral gyrus, bilateral supplementary motor area and right superior temporal gyrus) and VN (bilateral calcarine fissure and bilateral lingual gyrus) and short-range FCD in FPN (left middle frontal gyrus, left supplementary motor area and left superior frontal gyrus) and SMN areas (bilateral medial cingulate gyrus). Furthermore, the study highlighted that reductions in long-range FCD were primarily observed in the FPN area (including right inferior frontal gyrus, right inferior parietal lobule, right medial prefrontal cortex, right middle frontal gyrus, right superior frontal gyrus and right supramarginal gyrus) (Ding et al., 2014); see Table 3D.

3.6.2. Task-based functional MRI

Two studies conducted task-based functional MRI assessments in FS. In the first study on 12 FDS and 12 HC, a stress-inducing task revealed hyper-reactivity in the LN (including bilateral amygdala and left hippocampus) in FDS compared to HC (Allendorfer et al., 2019). The second study studied facial emotion processing differences between 12 FDS and 24 HC (Szaflarski et al., 2018). This revealed hyper-reactivity to happy faces, primarily in DAN (left superior frontal gyrus, left superior parietal lobule and right lateral occipital cortex) and other networks including VAN (right central opercular cortex), VN (left lateral occipital cortex), SMN (left postcentral gyrus), and DMN (right superior temporal gyrus/middle temporal gyrus). Similarly, neutral faces induced hyper-reactivity in the DA (left precentral gyrus, right superior frontal gyrus/middle frontal gyrus and right precentral gyrus/middle frontal gyrus) and SMN (left superior parietal lobule, left postcentral gyrus and left precentral gyrus). Moreover, the study reported hypo-reactivity to negative facial emotions, predominantly in the FPN (right angular gyrus/ superior parietal lobule, left cingulate gyrus and right frontal pole), followed by VN (right fusiform gyrus /lingual gyrus and left parahippocampal gyrus), VAN (left central opercular cortex and left parietal operculum cortex), DMN (left frontal opercular cortex/inferior frontal gyrus), SMN (right putamen), and cerebellum. Additionally, hypo-reactivity was observed to happy faces in the VAN (left frontal orbital cortex) and cerebellum, and to neutral faces in the VAN (right insula).

3.7. Meta-analysis

3.7.1. Meta-analysis of cortical thickness abnormalities

Pooling data from the seven cortical thickness studies in meta-analysis involved a total of 198 patients and 254 HC, identifying two significant clusters, individuals with FDS demonstrated reduced cortical thickness—the right calcarine fissure/surrounding cortex ($z = -3.08$, $p < 0.01$) and right precentral gyrus ($z = -3.11$, $p < 0.01$), using standard permutation tests with a threshold of uncorrected p-value < 0.005 and $|z| > 1$. At a less stringent threshold of < 0.05, individuals with FDS demonstrated reduced cortical thickness in two additional clusters, the left inferior frontal gyrus, triangular part ($z = -2.34$, $p < 0.01$) and right postcentral gyrus ($z = -2.11$, $p = 0.02$); see Table 4. It is important to note that no convergent increases in cortical thickness were observed.

3.7.2. Meta-analyses of whole-brain seed-based rsFC abnormalities

Following the categorization of seed ROIs into seven distinct seed-networks, three studies of the sensorimotor (48 FDS and 64 HC), three studies of the default mode (48 FDS and 64 HC) and two studies of the limbic seed-network (24 FDS and 36 HC) were meta-analysed. Since only one study contributed to the visual, dorsal attention, ventral attention, and fronto-parietal seed-networks, meta-analyses for these were not carried out (Eickhoff et al., 2012). The statistical significance of the ALE scores was assessed using a permutation test with 1000 iterations, applying cluster-level inference at a p-value threshold of less than 0.05 (FWE), with a cluster-forming threshold set at a p-value of less than 0.05. Attempts to increase the number of permutations (e.g., 5000) did not converge, potentially due to the very limited number of contributing studies ($n = 3$).

Sensitivity analysis, based on the number of contributing studies per cluster, revealed that only the sensorimotor and default mode seed-networks yielded significant results, and the limbic seed-network did not yield a cluster with contributions from at least two studies. The meta-analysis of the sensorimotor seed-network (17 foci from 3 experiments) revealed a significant peak cluster in the left thalamus ($z = 3.73$, $p < 0.01$) extending over the left putamen ($z = 3.62$, $p < 0.01$). This result illustrated increased FC of seeds located in the SMN in FDS participants compared to HC (Table 5A). In the meta-analysis of the default mode seed-network (12 foci from 3 experiments), a single significant cluster was identified, mapping to the left insula ($z = 4.01$, $p < 0.01$). This exhibit heightened FC with seeds located in the DMN in FDS participants compared to HC (Table 5B).

4. Discussion

This systematic review and meta-analysis aimed to address the existing knowledge gap by identifying spatially consistent brain regions associated with FDS by analysing quantitative structural and functional MRI studies. Our structural MRI meta-analysis revealed significant reductions in cortical thickness within the right calcarine fissure/surrounding cortex and the right precentral gyrus, indicating structural abnormalities in visual and sensorimotor areas in people with FS. Furthermore, the rsFC meta-analysis demonstrated increased connectivity between seeds in the SMN and left thalamus, extending to left putamen areas within the VAN and seeds in the DMN to the left insula, in FDS group. These findings highlight potential structural and functional neurobiological mechanisms associated with FS, consistent with our initial aims. The upcoming sections will detail the interpretation and implications of these findings in the context of existing literature on FDS and FND in general.

4.1. Structural imaging

Previous evidence indicates that FDS is associated with both increased and decreased cortical thickness (Mcsweeney et al., 2017; Bègue et al., 2019; Foroughi et al., 2020; Hernando et al., 2015; Hassan et al., 2024). This study, therefore, used anisotropic effect-size SDM to integrate both positive and negative findings to identify patterns of cortical thickness difference in FS (Radua and Mataix-Cols, 2012)—the first meta-analysis of morphometric features in FDS to the best of our knowledge. We found decreased cortical thickness in the calcarine fissure/surrounding cortex (Brodmann area 18) and the precentral gyrus in FS, with no regions showing convergent increases.

The calcarine fissure and surrounding cortex contain the visual area, responsible for processing visual information received from the eyes (Grill-Spector and Malach, 2004). Structural changes such as reduced cortical thickness in this region may be linked to visual disturbances sometimes reported by individuals with FS, including blurred or distorted vision, visual hallucinations, or even transient blindness (Lenka et al., 2015). Furthermore, the visual cortex is part of a larger network that integrates sensory information to form a comprehensive representation of the body in space. This integration is critical for spatial awareness, coordination, and balance and disruptions can potentially lead to issues such as derealization, depersonalization, or balance problems, which are features that can be seen in FDS patients (Lamme et al., 2000). The precentral gyrus serves as the primary motor cortex and is crucial for planning and executing voluntary movements (Bhattacharjee et al., 2021). Abnormalities, such as reduced cortical thickness, might be associated with impaired motor control, leading to weakness, involuntary movements or incoordination in FDS individuals (Pakkenberg and Gundersen, 1997). Additionally, the precentral gyrus plays a role in sensory integration with motor commands (Avanzino et al., 2015). Structural abnormalities could affect this integration, resulting in altered perception of movement or body position, potentially causing motor symptoms like gait disturbances or balance problems, observed in individuals experiencing FS. It is crucial to recognize that these areas do not function in isolation. They are parts of larger neural networks, and dysfunction in one area can affect others. The observed abnormalities could be just one piece of a broader puzzle reflecting changes in network connectivity. For example, altered precentral gyrus activity might influence networks associated with body awareness or sensory processing within the visual cortex, and vice versa (Pouget and Sejnowski, 1997; Papadelis et al., 2016).

Several neuroimaging studies have reported morphological deviations in the sensorimotor cortex and visual areas related to other forms of FND, as well as correlations between neuroimaging changes in these regions and clinical features in individuals with FS. One study found a significant positive correlation between the thickness of the right lateral occipital cortex and depersonalization/derealization scores in individuals with FND (Perez et al., 2018a). Additionally, a VBM study on functional movement disorders identified increased grey matter in the subcortical areas (including thalamus, amygdala and striatum), fusiform gyrus, and cerebellum, along with reduced volumes in the sensorimotor cortex compared to HC (Maurer et al., 2018). Furthermore, another study comparing patients with FDS to individuals with syncope reported that longer illness duration was associated with reduced cortical thickness in several brain clusters, including the left postcentral gyrus, right pericalcarine cortex, left temporoparietal junction, and various midline structures. Similarly, longer duration of dissociative seizures correlated with decreased cortical thickness in several cortical area including the left precentral gyrus, lateral occipital cortex, etc (Zelinski et al., 2022).

FS/FND have been linked to various etiological factors, including proximal trauma and early life maltreatment (ELM) (Ludwig et al., 2018; Kanaan and Craig, 2019). Morphological studies have identified differences in specific brain regions, including the sensorimotor and visual areas, related to ELM. Tomoda *et al.* found childhood sexual abuse was linked to reductions in grey matter volume in both the right and left visual cortex, as well as thinning of the lingual gyrus, left fusiform gyrus and middle occipital gyrus (Tomoda et al., 2009). Heim *et al.* reported cortical thickness decreases in parts of the sensorimotor cortex, including those mapping the clitoris and extended genital regions, in those suffering childhood sexual abuse (Heim et al., 2013). It has also been reported that witnessing multiple episodes of domestic violence during childhood has been associated with reduced grey matter volume in the right lingual gyrus, left occipital pole, and bilateral secondary visual cortex (Tomoda et al., 2012). Based on another study, individuals who witnessed or experienced parental domestic violence in early childhood exhibited reduced FA in the left inferior longitudinal fasciculus (Choi et al., 2012b). This tract serves as a visual-limbic pathway, facilitate transmission of visual information between the occipital and temporal regions, which are critical for memory, learning, and emotional processing (Catani, 2003). Although there is substantial evidence linking ELM to structural and functional brain changes, longitudinal studies with and without ELM exposure remain limited. This makes it difficult to determine whether the observed morphological changes reflect the direct effects of ELM, an underlying brain vulnerability, or an interaction of these factors. Moreover, we were unable to directly assess the specific effects of ELM in FDS subjects within this study due to lack of data. However, recent imaging studies indicates that changes in the grey matter structure of the visual regions and the sensorimotor cortex in individuals with FDS could be influenced by ELM experiences. Furthermore, our findings emphasize the potential importance of occipital and visual regions in FDS, particularly given their associations with early life maltreatment and dissociative experiences. This pattern may be as crucial as the traditionally highlighted limbic and hippocampal abnormalities, suggesting that abnormalities in visual and body representation systems play a more prominent role in the neurobiology of FDS than previously recognized.

Notably, while our meta-analysis identified regions of decreased cortical thickness, no convergent increases emerged. This asymmetry (decreases without increases) may indicate that cortical thinning, particularly in visual and sensorimotor cortices, represents the more reproducible morphometric signal in FDS. Alternatively, the absence of convergent thickening could reflect limited statistical power, which may disperse potential effects and prevent cross-study alignment. Accordingly, these findings should be viewed as preliminary patterns rather than definitive markers until future harmonized, higher-powered studies are done.

4.2. Functional imaging

Available studies on FS/FND underscore the role of aberrant FC in the pathophysiology of this disorder, though significant inconsistencies among studies have been noted (Mcsweeney et al., 2017; Foroughi et al., 2020; Hassan et al., 2024; Perez et al., 2018b; Sasikumar and Strafella, 2021). We therefore conducted a coordinate-based meta-analysis of seed-based, resting-state FC studies to identify convergent patterns of aberrant connectivity of large-scale brain networks in individuals with FS. To our knowledge, this meta-analysis is the first to integrate such diverse findings, providing evidence of significant hyperconnectivity between the SMN-VAN and DMN-SMN in FDS patients. In addition to the positive findings, we found no consistent clusters for the limbic seed-network, and convergence could not be evaluated for visual, dorsal attention, ventral attention, or fronto-parietal seed-networks because only one study contributed to each. These null results suggest that abnormalities are unlikely to be uniform across all brain systems and may reflect both true negatives and limited power/methodological heterogeneity.

The pathophysiology of FS/FND involves complex interactions between multiple brain networks (Perez et al., 2021) and previous studies highlighted the aberrant connectivity within and between SMN, DMN and VAN in FND patients (Lo et al., 2025). A resting-state fMRI study, comparing motor-FND patients to HC found increased FC between motor regions to the VAN/salience network (including bilateral posterior insula, temporoparietal junction, middle cingulate cortex, and putamen) (Diez et al., 2019). Another study on motor-FND has documented VAN and SMN hyperactivity during emotional processing tasks (Aybek et al., 2015). Moreover, FND patients including FDS exhibited increased frequencies of insular co-activation patterns related to salience network, SMN, and DMN compared to HC (Weber et al., 2024). Furthermore, a study on children with FND (including FS) has demonstrated significant connectivity abnormalities across various brain networks, including the SMN, salience network, FPN, VN, DAN, cerebellum, and language networks (Rai et al., 2022). Notably, similar patterns of aberrant connectivity have also been observed in other functional somatic syndromes, such as functional constipation, where increased FC between the salience network and SMN has been reported (Duan et al., 2021). Moreover, our findings can be interpreted within the framework of disruptions in predictive coding in FND (Hallett et al., 2022a). From a predictive processing perspective, reduced cortical thickness in visual and sensorimotor areas may reflect abnormalities in the precision weighting of sensory signals, while hyperconnectivity between SMN, DMN, and VAN could indicate maladaptive integration of top-down expectations with bottom-up inputs. Such disruptions could underlie core FS symptoms, including abnormal motor responses and altered bodily self-awareness. Using predictive processing as a framework helps unify structural and functional findings, suggesting that FS may emerge from aberrant attempts to resolve persistent prediction errors across sensory and motor systems (Friston, 2010; Dohmatob et al., 2020; Spratling, 2008; Palaniyappan and Liddle, 2012).

The high prevalence of comorbid psychiatric disorders associated with FDS warrants consideration of its neurobiological underpinnings. A systematic review and meta-analysis has revealed increased FC between the SMN and both the VAN and DMN in PTSD compared to HC (Bao et al., 2021). Furthermore, hyperconnectivity between the SMN and attention network, as well as between the DMN and SMN associated with PTSD. In depression, another common comorbidity in FS, a study showed significantly increased FC within the VAN and between the VAN and SMN, which was positively correlating with both Hamilton Depression Rating Scale and Childhood Trauma Questionnaire scores (Cui et al., 2024). Furthermore, increased SMN-VAN FC has been linked to depression (Yu et al., 2019; Mao et al., 2020). In addition, prolonged psychological stress exposure has been associated with increased activation in the DMN, DAN, VAN, SMN, and primary VN compared to controls (Soares et al., 2013). Further research is therefore crucial to disentangle whether the connectivity changes in FDS are a consequence of FDS or secondary to the effects of comorbid psychiatric disorders.

5. Limitations and future directions

This review and meta-analysis should be interpreted in light of several limitations. Firstly, our work included a limited number of studies, each with small sample sizes, which limits generalizability and may introduce bias into the data. Moreover, all included studies were rated as having a fair methodological quality, with none meeting the criteria for a good rating. This could limit the robustness of the findings, as we cannot be certain whether higher-quality studies would replicate or refine the patterns observed here. Future research with larger sample sizes and stronger study designs will therefore be essential to establish the reliability and generalizability of these results.

Secondly, we restricted inclusion to studies of patients with pure FDS, excluding mixed FND presentations (e.g., motor dysfunction co-occurring with FDS) unless FDS data were reported separately. We also excluded studies in which FDS occurred in the context of comorbid epilepsy. While these decisions reduced heterogeneity and ensured that our findings specifically reflect FDS, they may also limit generalizability to the broader FND population and to patients with comorbid epilepsy. Future studies with larger and more diverse cohorts are needed to clarify the neurobiological similarities and distinctions between FDS, other FND subtypes, and presentations complicated by comorbid epilepsy.

Thirdly, in line with standard practice, we synthesized coordinate-based data rather than integrating raw data from individual studies, which may reduce the precision and completeness of spatial effects. Future work adopting mega-analytic approaches would allow more comprehensive data integration. Moreover, our functional meta-analysis was based on only three contributing experiments. Simulation studies indicate that such small study pools undermine stability regardless of permutation count, with robustness driven primarily by the number of available experiments (Frahm et al., 2023). Thus, although we applied 1000 permutations in line with common practice, these findings should be considered preliminary until more studies are available.

Fourthly, although grouping ROIs into functional networks improved comparability across heterogeneous seed definitions, collapsing seeds into broad networks risks obscuring the distinct roles of individual subregions. As such, our findings should be interpreted as reflecting convergent network-level patterns rather than precise regional effects. Future studies using harmonized ROI definitions and selecting seeds across multiple networks will be important to clarify connectivity profiles more comprehensively.

Fifthly, we did not evaluate the impact of FDS semiology, comorbidities, or medication use due to the limited number of studies and available information. Future research should include subgroup analyses to provide deeper insights.

Sixthly, as our analyses focused on case-control designs, we cannot draw conclusions about causality or aetiology. Longitudinal studies will be crucial to determine how structural and functional alterations evolve over time, particularly in relation to treatment response and clinical recovery.

Finally, we encountered difficulties obtaining coordinates from all relevant studies despite our efforts to contact the authors, and four studies in the review and two in the functional meta-analysis used overlapping samples, which should be considered when interpreting the results.

In summary, to move the field forward, future studies should not only expand in sample size but also adopt harmonized, multi-center designs to minimize methodological heterogeneity. Longitudinal imaging will be essential to determine whether structural and functional abnormalities precede clinical onset, change with symptom course, or normalize with treatment. Subgroup analyses examining seizure semiology, trauma history, and psychiatric comorbidities could clarify sources of heterogeneity. Moreover, given that the directionality of the observed hyperconnectivity remains unclear, future studies should employ advanced analytical methods such as dynamic causal modeling in large sample sizes to elucidate causal interactions and temporal dynamics between these networks. In addition, pooling raw neuroimaging data across centers in mega-analyses will improve statistical power and allow more precise characterization of affected regions and networks.

6. Conclusion

Overall, our meta-analyses suggest that FDS manifestations may be associated with structural and functional abnormalities in visual, sensorimotor, attention and default mode areas and networks. These findings provide preliminary evidence of potential neurobiological mechanisms but require replication in larger, harmonized studies before firm conclusions can be drawn. Notably, the involvement of occipital and visual regions observed here highlights a novel aspect of FDS pathophysiology that warrants greater emphasis, along with limbic and other regions. Future research should integrate psychiatric assessments and other clinical features with neuroimaging parameters to better understand the neurobiological basis of FDS susceptibility.

Declaration of Competing Interest

The authors declare that they have no known competing interests.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.neubiorev.2025.106439.

Data availability

Data will be made available on request.

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